

DATA COLLECTION & AUTOMATION

OTel & HEC

What They Are, Why You Need Them, and Why Automate with Terraform



Agenda

What we'll cover in this session

01

What is OTel?

The OpenTelemetry Collector, and the role it plays

02

Why OTel is Needed

The problem it solves versus multiple vendor agents

03

What is HEC?

Splunk's HTTP Event Collector and how data reaches it

04

Why HEC is Needed

Why a token-based HTTP path matters for ingestion

05

OTel + HEC Together

How the two combine in a single data pipeline

06

Why Automate with Terraform

The case for infrastructure as code over manual setup

What is OTel?

A vendor-neutral standard for collecting telemetry data

OpenTelemetry (OTel) is an open-source standard and toolset for collecting metrics, traces, and logs from applications and infrastructure. The Splunk Distribution of the OpenTelemetry Collector runs as an agent or gateway, gathering telemetry and forwarding it to Splunk Observability Cloud — without locking you into a single vendor's proprietary agent.



One collector, one config format, many possible destinations — that's the core idea behind OTel.

Why OTel is Needed

The problem it solves compared to fragmented, vendor-specific agents

WITHOUT OTEL

- A separate agent per vendor, each with its own config format
- Switching monitoring tools means re-instrumenting everything
- Inconsistent metric names and tagging across teams
- More agents running on every host, consuming more resources
- Data locked into a single vendor's collection format

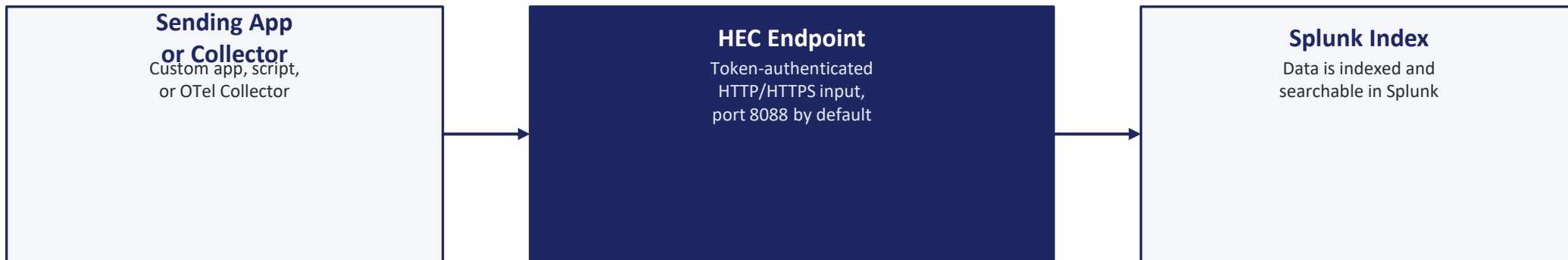
WITH OTEL

- One collector, one config format, for metrics, traces and logs
- Change backends without touching application instrumentation
- Consistent, standardized telemetry across every team
- A single lightweight process instead of several agents
- You own your data pipeline — no vendor lock-in

What is HEC?

Splunk's token-based HTTP API for sending data in

The HTTP Event Collector (HEC) is Splunk's method for sending data and events into a Splunk platform deployment over HTTP or HTTPS. It uses token-based authentication: an application, script, or the OTel Collector sends data to a HEC endpoint along with a token, and Splunk indexes it — no separate forwarder installation required on the sending side.



HEC is firewall-friendly and doesn't require deploying a Splunk forwarder on the source system.

Why HEC is Needed

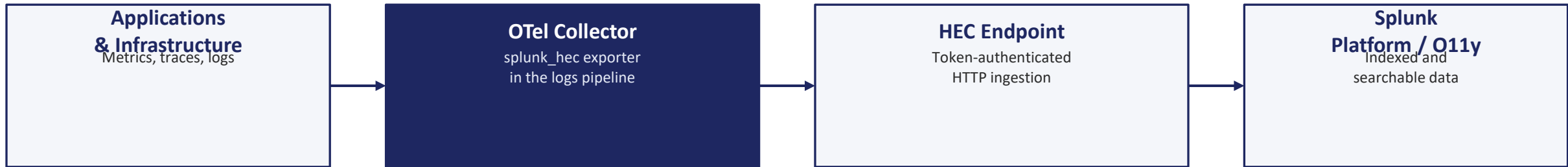
Four outcomes teams get once HEC is in place

No Forwarder Required	Token-Based Control	Firewall Friendly	Flexible Routing
Send data straight from an app, script, or the Collector over HTTP	Each token can be scoped to a source, sourcetype and index	Standard HTTP/HTTPS traffic, easy to allow through network policy	Split metrics, logs and traces to the destinations that need them

Together, these make HEC the simplest bridge between custom or third-party systems and Splunk.

OTel + HEC Together

How the Collector uses HEC as one of its export paths



The Collector standardizes how data is gathered; HEC is one of the paths it can use to deliver that data to Splunk.

Why Automate with Terraform

The problems teams hit once Collector and HEC setup is done by hand

MANUAL SETUP

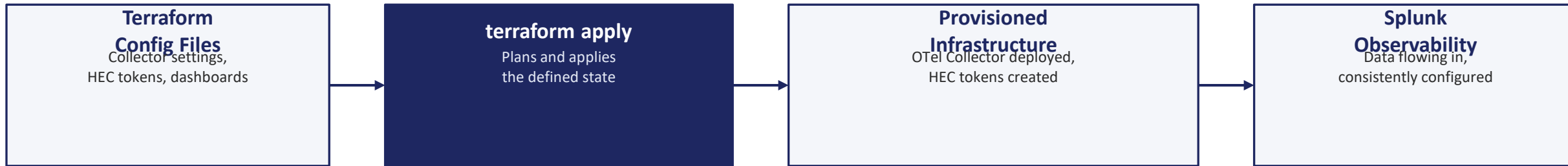
- Every host or cluster configured by hand, one at a time
- Config drift between environments is hard to detect
- No history of what changed, when, or why
- Re-deploying after a failure is slow and error-prone
- Access tokens and settings often live only in someone's memory

TERRAFORM AUTOMATION

- Same Collector and HEC config applied consistently everywhere
- Environments stay identical because they're defined in code
- Every change is version-controlled and reviewable
- Rebuild or recover an environment in minutes, not hours
- Tokens and settings are managed centrally, not by memory

How Terraform Automation Works

From code to a fully configured observability pipeline



The same configuration can be applied to a new environment, a disaster-recovery region, or an entire fleet — with one command.

Key Takeaways

What to remember from this session

OTel

A vendor-neutral standard collector for metrics, traces and logs — one agent, many backends

HEC

Splunk's token-based HTTP path for sending data in, no forwarder installation required

Together

The Collector uses HEC as one export path into the Splunk platform

Terraform

Turns Collector and HEC setup into consistent, version-controlled, repeatable code

Bottom line: standardize collection with OTel, deliver it reliably with HEC, and keep it consistent with Terraform.